Usability Engineering

UE 3 · Evaluation

Frank Viktoria | Freller Lisa-Maria | Keck Mandy | Neumayr Thomas

OVERVIEW

UE1 | Get a topic and brainstorm ideas for digital product → **chapter 1 2 3**

- Brainstorm & cluster your ideas
- Create 3 scenarios
- Homework: Storyboards

UE2 | Sketch UI & Prototyping Low-Fidelity → **chapter 4 5**

- Wireframes, clickable grayscale prototype
- Homework: Finish clickable prototype → **present UE3**

UE3 | Heuristic Evaluation & present prototype → **chapter 6 7**

- Prototyping High-Fidelity „Redesign“
- Homework: Finish High-Fidelity Prototype

UE4 | Preparation for Usability Testing (all groups) → **chapter 8**

- 2 test scenarios & 6 tasks per scenario
- Pretest + Usability Test
- Homework: Analysis of UT & redesign suggestions



UX LIFECYCLE

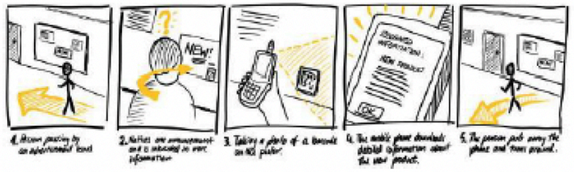
Aiming to meet user needs and requirements

Requirement Analysis

Scenarios, Personas

Study

Design



Sketching

Scribbles, Storyboards



Image by Marli Ritter

Evaluate

Prototype

(GUI) Design

GUI Sketches, Wireframes



Evaluation

Heuristic Evaluation, Usability Tests,..

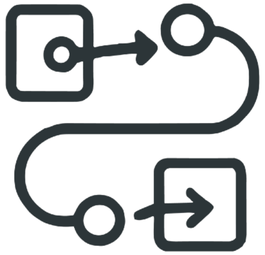
Develop

Coding,...

Building and Prototyping

[Hartson & Pyla, 2019]

TODAY'S LESSON



HEURISTIC EVALUATION

Evaluate low-fi prototype of other group



PRESENTATION

5 minutes <> group

Focus on idea & low-fi prototype



HIGH-FI PROTOTYPE

Create high-fi prototype in figma

HEURISTIC EVALUATION

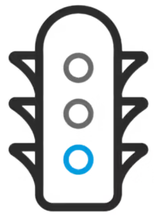
Overview

- It's an **inspection** method – inspect a design to find usability problems or future problems of users
- Efficient, fast and inexpensive method to identify usability problems
- **Experts** (sometimes also developers) check if design is in line with design principles or design guidelines
- Based on HCI researchers experience distilled into usability principles (=heuristics) which represent outcomes of numerous user studies
- Can be used intuitively (after a brief training period) compared to other evaluation methods
- Outcome: **List of Design-Problems** ordered by problem severity

HEURISTIC EVALUATION

10 Heuristics Nielsen

General principles called "heuristics" because they are broad rules of thumb and not specific guidelines



Visibility of System Status



Match Between System & the Real World



User Control & Freedom



Consistency & Standards



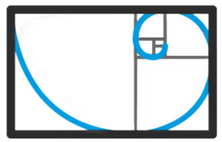
Error Prevention



Recognition Rather than Recall



Flexibility & Efficiency of Use



Aesthetic & Minimalist Design



Help Users Recognize, Diagnose & Recover from Errors



Help & Documentation

Img Source: https://www.interaction-design.org/literature/topics/heuristic-evaluation
Short video summary by David Lazarus: https://www.youtube.com/watch?v=hWc0Fd2AS3s

HEURISTIC EVALUATION

10 Heuristics Nielsen

1. Visibility of system status: The system should at all times inform the user what it is doing. The system should respond with an appropriate answer (i.e. understandable) in a reasonable time (i.e. within a few seconds).

2. Match between system and the real world: The system should speak the language of the user, using words, phrases, and concepts that are familiar to the user. Less system-oriented messages, especially codes, should be used. In addition, the conventions of the real world should be taken into account. The information or messages must be displayed in a natural and logical order.

3. User control and freedom: The user makes mistakes when operating the system, i.e. when selecting system functions. Then he needs a clearly marked emergency exit to undo the undesired selection. He should be able to do this without having to go through complex dialogs. Functions such as undo and redo should be supported by software.

Source: https://www.nngroup.com/articles/ten-usability-heuristics/

HEURISTIC EVALUATION

10 Heuristics Nielsen

4. consistency and standards: The user should not stumble over different terminologies, situations or actions that actually describe the same thing. An application should therefore follow the platform conventions and present the same facts in a uniform way.

5. Error prevention: A prudent design that prevents errors before they occur is always better than a good error message. Therefore, ambiguity and confusion should be avoided and unimportant information should be hidden.

6. Recognizing rather than recalling: Objects, actions and options should be visible, i.e. available on the user interface. The user should not have to remember every dialog sequence on how to reach instructions for using the system. They should be accessible whenever they are needed.

7. Flexibility and efficiency of use: Accelerators or shortcuts (for example, icons in toolbars or keyboard shortcuts) that are not seen or needed by the novice user can often increase the speed of use for experienced users. The user should also be able to design them himself.

Source: https://www.nngroup.com/articles/ten-usability-heuristics/

HEURISTIC EVALUATION

10 Heuristics Nielsen

8. Aesthetic and minimalist design: Dialogs should not contain information that is unimportant or unusual. Any additional information in a dialog competes with the relevant information and reduces its relative visibility. Information should always be as brief and substantive as possible, i.e. any unnecessary information should be avoided.

9. Help users recognize, diagnose, and recover from errors: Error messages should be in simple language. No codes or messages incomprehensible to the user should be issued. They should precisely describe the cause of the problem and constructively indicate a solution path. The easiest way to solve the problem is to undo the error.

10. Help and documentation: In any case, it is better if the system can be used without documentation. Help and documentation can be provided. In this case, any information should be easy to search and find. The help should be focused on the task the user is currently accomplishing. A concrete list of steps should be given, which describes a solution path for the task and leads the user to the goal as quickly as possible.

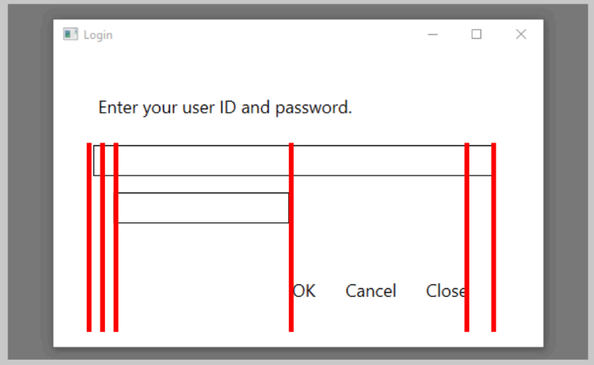
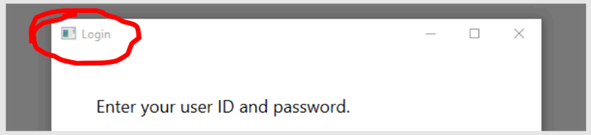
Source: https://www.nngroup.com/articles/ten-usability-heuristics/

HEURISTIC EVALUATION

Practical implications

- Heuristic evaluations last 1 – 2 hours
- Each evaluator does a heuristic evaluation of an interface alone
- Sometimes an observer can record the evaluator's comments, sometimes the evaluator does
- Evaluators can be given scenarios that describe typical usage scenarios (built from a task analysis of users)
- **Each expert** goes through the interface several times
  1. to get a feel for the scope of the system
  2. to focus on specific elements
- Evaluators produce a **list of usability problems**: the usability principle and the design feature, that violated it

HEURISTIC EVALUATION

No.	Issue Title	Issue Description	Conflicted Heuristic	Screenshot	Severity
3	Falsches Layout	Die Label und die Inputfelder haben unterschiedliche x-Positionen und Breiten	Alignment, Aesthetic & minimalist Design		2
4	Nur „Login“ als Titel	Aus „Login“ lässt sich nicht darauf schließen, wo man sich gerade einloggt.	Visibility of system status, Error Prevention		2

The following 0-4 rating can be used to rate the severity of usability problems:

0 = I don't agree that this is a usability problem at all

1 = Cosmetic problem only: need not be fixed unless extra time is available on project

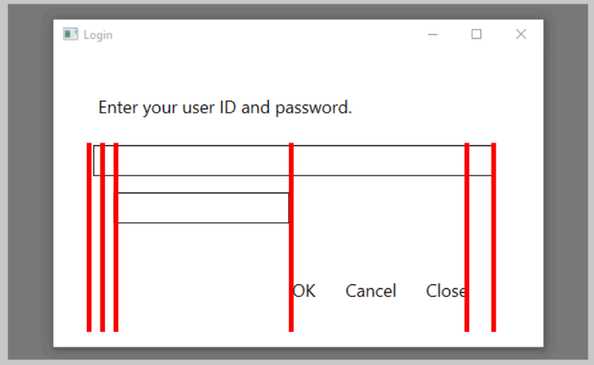
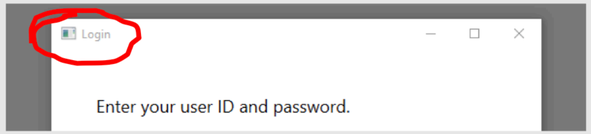
2 = Minor usability problem: fixing this should be given low priority

3 = Major usability problem: important to fix, so should be given high priority

4 = Usability catastrophe: imperative to fix this before product can be released

Source: https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/

HEURISTIC EVALUATION

No.	Issue Title	Issue Description	Conflicted Heuristic	Screenshot	Severity
3	Falsches Layout	Die Label und die Inputfelder haben unterschiedliche x-Positionen und Breiten	Alignment, Aesthetic & minimalist Design		2
4	Nur „Login“ als Titel	Aus „Login“ lässt sich nicht darauf schließen, wo man sich gerade einloggt.	Visibility of system status, Error Prevention		2

The severity of a usability problem is composed of the following 3 factors:

- The frequency of occurrence of a problem (how often/rarely does it occur?).
- The impact of the problem when it occurs (is it easy or difficult for users to solve the problem?)
- The sustainability of the problem (is it a one-time problem that can be easily solved or are users permanently disturbed by the problem)?

Source: https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/

Group	Evaluates Group
A – E – I	B – F – J
B – F – J	C – G – K
C – G – K	D – H – L
D – H – L	A – E – I

UG1 – UG2 – UG3

HANDS ON 👋

1 | **Heuristic Evaluation** for another team

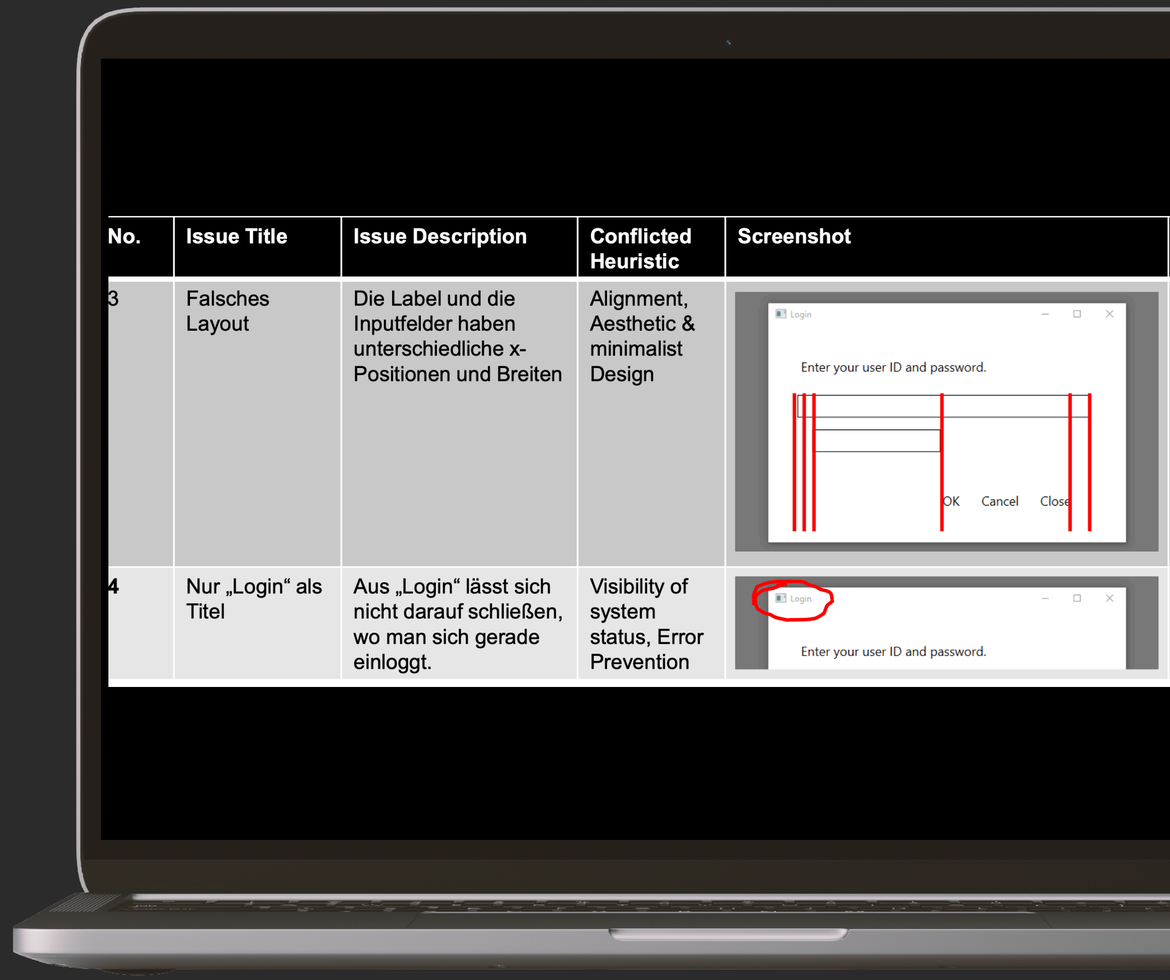
Individual

2 | Merge results, then submit results group you evaluated

Group

3 | **Presentation**

4 | Start **High-Fi Prototype**



No.	Issue Title	Issue Description	Conflicted Heuristic	Screenshot
3	Falsches Layout	Die Label und die Inputfelder haben unterschiedliche x-Positionen und Breiten	Alignment, Aesthetic & minimalist Design	
4	Nur „Login“ als Titel	Aus „Login“ lässt sich nicht darauf schließen, wo man sich gerade einloggt.	Visibility of system status, Error Prevention	

PROMPTING & AI

Teams need to understand & evaluate AI & write „effective“ prompts

What is still needed:

✓ Design skills

✓ Domain knowledge

✓ Creativity



👉 AI doesn't replace design, but it changes the way we work

Source: https://www.figma.com/de-de/reports/design-systems-power-the-pace

FIGMA MAKE

AI-driven tool that turns ideas into prototypes

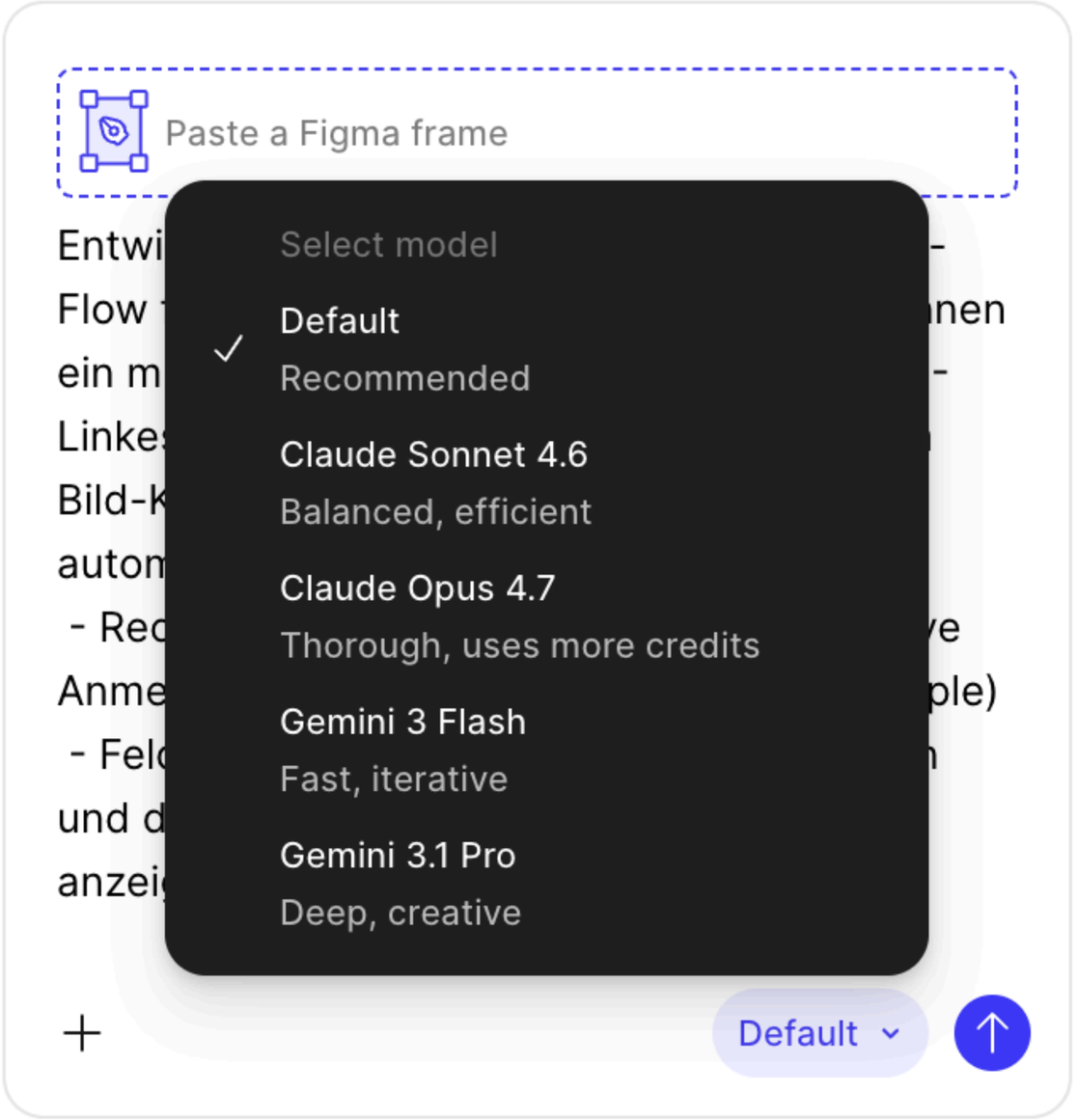
Pitfalls:

✓ Similar, flat, or repetitive designs

✓ Limited effectiveness in complex domains

✓ Weak design system integration

✓ Design-to-code gap: generated code not production-ready

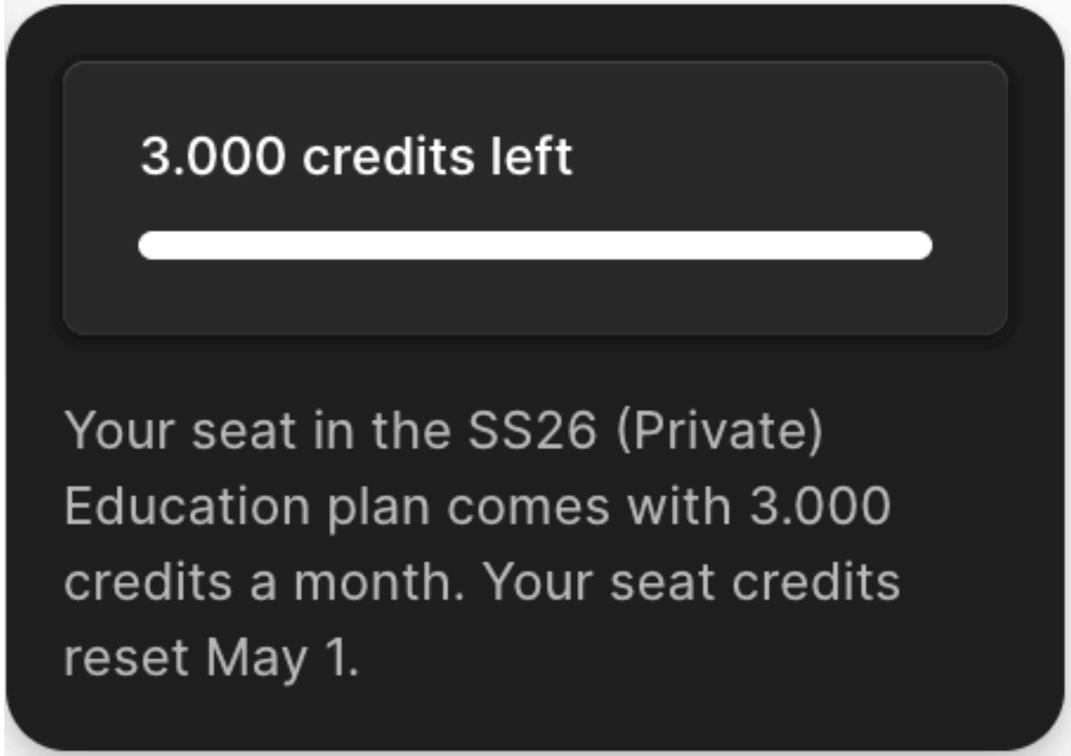


Source: https://www.figma.com/de-de/reports/design-systems-power-the-pace

PROMPTING

think 💡 before you prompt

- Credit: even simple changes (e.g. button color) can cost up to 65 credits depending on model
- Switching between models helps save credits
- Create templates to avoid repeated credit usage



Source: https://www.figma.com/de-de/reports/design-systems-power-the-pace

act as persona

bullet points for readability

add what not to do

attach visual reference/library

first prompt is important, then iterate

Act as an experienced UX designer.
Design a smartwatch interface for a fitness app.

Requirements

- Use a minimal, clean design style
- Optimize for small screen usability and readability
- Include a bottom navigation bar with 3–4 core sections (e.g. Home, Activity)
- Focus on clear hierarchy, large touch targets, and simple interactions
- Use concise labels and recognizable icons
- Do not use gradients

Content

- Display key fitness data (e.g. steps, heart rate, calories)
- Include one primary action (e.g. “Start Workout”)

Style

- Minimal color palette (2–3 colors max)
- High contrast for accessibility
- Use spacing and typography to create clarity

Design System Thinking

- Use consistent spacing, components, and icon style
- Structure the UI in reusable components (cards, metric blocks, buttons)
- Ensure visual consistency across all screens

HANDS ON 👋

- Use Figma Make to redesign 1–2 screens from your wireframes and develop a visual design for your high-fidelity prototype.
- Reflect on the following questions about the Figma Make output:
  - **Prompts as a design brief:** What prompts did you use? What did you need to state explicitly in the prompt that you would normally know or assume implicitly when designing yourself?
  - **Evaluating the output:** Did the result meet your expectations? When was the AI-generated output “good enough”? Where did you need to revise or iterate? What did Figma Make misunderstand? Which elements did you keep for the final design?

SUBMISSION

Homework



- Hand in merged evaluation results you created as a group from the group you evaluated
- Semester documentation:
  - **Chapter 6** Heuristic Evaluation
    → include HE results by the other group showing your prototype
  - **Chapter 7** High-Fi Prototype

REFERENCES

Buxton B., 2007. Sketching User Experiences: Getting the Design Right and the Right Design. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.

Greenberg, S. & Carpendale, S. & Marquardt, N. & Buxton, B. (2012). Sketching User Experiences. 10.1016/C2009-0-61147-8.

Hartson, R. & Pyla, P. (2019). The UX book: Agile UX design for a quality user experience, 2nd edn, Elsevier Inc, Cambridge, MA, Science Direct

Moser, C., User Experience Design: Mit Erlebniszentrierter Softwareentwicklung zu Produkten die Begeistern, Springer Verlag, 2012

REFERENCES

Nielsen J., *10 Usability Heuristics for User Interface Design*. 1995. url: http://www.nngroup.com/articles/ten-usability-heuristics/ (besucht am 28. 11. 2022)

Nielsen J., *Usability Engineering*. Morgan Kaufmann, 1994

Nielsen J.. „Usability Inspection Methods". In: *Conference Companion on Human Factors in Computing Systems*. CHI '94. Boston, Massachusetts, USA: ACM, 1994, S. 413–414. url: http://doi.acm.org/10.1145/259963.260531

Nielsen J. und Mack R. L., *Usability Inspection Methods*. New York, NY, USA: John Wiley und Sons, 1994

Nielsen J und Molich R., „Heuristic Evaluation of User Interfaces". In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. CHI '90. Seattle, Washington, USA: ACM, 1990, S. 249–256

Sauro J.. *How effective are heuristic evaluations?* Techn. Ber. Denver, Colorado: MeasuringU, Sep. 2012. url: https://measuringu.com/effective-he/

Smith S., RMA-Consulting http://de.slideshare.net/pubsmith/sketching-interfaces-workshop-interactions12-dublin